



Wastewater Business Plan

Lift · ATU · WWTP · Phase 1 Cloud · 2026

A company powered by **The Purple Standard**
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Vertical business plan · Version 1.3

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Phase 1 SaaS · 500-device fleet · cellular cost model

Executive Summary

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mooreVIEW — Wastewater Business Plan

Lift stations · ATU · WWTP · Phase 1 Cloud SaaS · 2026

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Executive summary

mooreVIEW enters the **US wastewater and onsite treatment market** as a **field-service monitoring platform** — proactive lift-station, ATU/ATS, and WWTP head-end monitoring with portable `.est` projects, PdM → CMMS, and MQTT Parc fleet uplink to **Phase 1 Cloud SaaS**.

Metric	Year 1	Year 2	Year 3
Total revenue (directional)	~\$550K	~\$1.74M	~\$3.9M
Cumulative monitored assets	~230	~380	~500+
End-of-year renewal ARR	~\$55K	~\$175K	~\$400K

Renewal sensitivity (\$20 / \$25 / \$30 per mo @ 500 assets): **\$120K – \$180K/yr** uniform commercial — see Part 4.

Overhead (OH): \$3/device/mo in Year 1 (G&A, support, ops — scales with fleet). **Year 2+: 7% of total revenue** (replaces per-device OH as the business scales).

Metric	Year 1	Year 2	Year 3
Total revenue	~\$550K	~\$1.74M	~\$3.9M
Infrastructure (cloud + cell cost)	~\$9K	~\$15K	~\$19K
Overhead	~\$6K (\$3/dev/mo)	~\$122K (7%)	~\$273K (7%)
Net after infra + OH	~\$535K	~\$1.60M	~\$3.61M

Y1 OH ≈ \$3 × ~170 avg fleet-months. Y2–Y3 OH = 7% × total revenue (install + renewal + cell).

Phase 1 cloud stack (SaaS + MQTT + Archive + Mongo) supports the full **500-device wastewater fleet** on a single NYC1 topology. At steady state:

Category	Monthly	Annual
Renewal revenue (500 assets, list)	~\$10,750	~\$129,000
DO Phase 1 cloud stack	~\$97	~\$1,167
Cellular — 500 field assets (\$2 SIM + \$0.50/50 MB)	\$1,250	\$15,000

Cellular — 32 WWTP/head-end appliances (\$2 SIM + \$5/1 GB)	\$224	\$2,688
Total infrastructure	~\$1,571	~\$18,855
Contribution after infra	~\$9,179	~\$110,145
Infra as % of renewal revenue	~14.6%	~14.6%

Steady-state table is **renewal + cell** on 500 assets. Company OH in Y3 (7% of ~\$3.9M total) is ~\$273K/yr — see Part 4 P&L.

Cloud alone is ~0.9% of renewal revenue at 500 assets. **Cellular backhaul** is the dominant opex line (~11.6% of renewal at cost; billed to customer at **\$3/mo** list = **\$1,500/mo** revenue vs **\$1,250/mo** cost → **\$250/mo** cell margin).

Serviceable market: ~\$400M/yr US wedge (sites without full SCADA). **5-year ARR ceiling (wastewater vertical):** \$15 – 25M recurring.

Beachhead channel: Purple Standard sister operators (**ACE Septic & Waste**) — reference fleet, install crews, first **100–500 sites** without cold national sales.

Part 1 — Market & product

Vertical scope

Segment	Asset	Proactive wedge	Project pack	Tier
Lift stations	Duplex/triplex pump stations	Level/pump trends before SSO	duplex-lift-station	Commercial
ATU / ATS	Residential & commercial treatment	Blower/UV/pump health; compliance route	ATS Compliance Pack	Res / Com
WWTP head-end	Small municipal / package plants	Pump/blower PdM; remote operator view	mle-wastewater	Commercial
Septic OEM	Panel-builder embed	Provider fleet Parc	ATS.est	OEM

Who buys

Buyer	Wastewater use case
Service contractors (ACE-class)	Monitoring contracts on lift + ATU routes
Utilities / collections districts	Fleet Parc; fewer SSO events; 13 mo compliance archive
OEM panel shops	Connected ATS panel vs dialer-only skid
Property managers / commercial	ATU compliance before inspection failure

vs legacy

Solution	Year-1 TCO (duplex lift)	PdM → CMMS
Autodialer	~\$600 – \$1,200	○
mooreVIEW (+ cell)	~\$2,836	●
VTScada (1K I/O)	~\$7K – \$12K	○

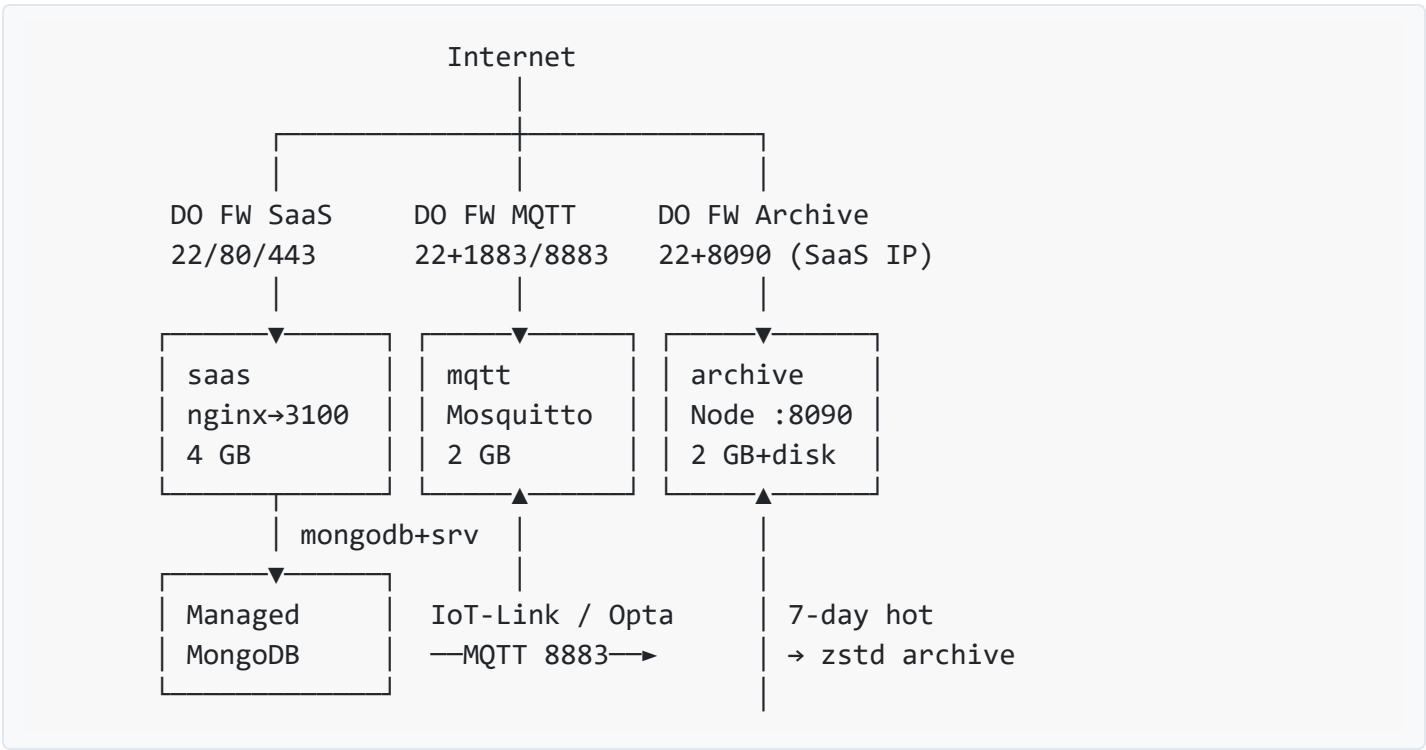
List pricing (wastewater assets)

Asset	One-time	Renewal/mo	Cell (list)	Year-1 total
ATU — residential	\$1,500	\$18	\$3	\$1,752
ATU — commercial	\$2,500	\$25	\$3	\$2,836
Lift station — duplex	\$2,500	\$25	\$3	\$2,836
Well & pump (adjacent)	\$2,500	\$25	\$3	\$2,836

Renewal includes Parc uplink, historian, CMMS, PdM, alarm profiles, platform updates.

Part 2 — Phase 1 Cloud SaaS platform

Production topology (NYC1)



Host	Role	Size
mooreview-saas	nginx → Cloud Studio :3100	4 GB / 2 vCPU
mooreview-mqtt	Mosquitto TLS :8883	2 GB / 1 vCPU
mooreview-archive	Cold historian API :8090	2 GB + block volume
Managed MongoDB	Hot historian (7-day rolling)	10–20 GB tier

Edge path: IoT-Link or Arduino Opta (cellular) → `mqtt://mqtt.mooreview.io:8883` → SaaS Parc ingest → Mongo hot → day-7 zstd export to archive.

WWTP head-ends (32 units): MVP Suite appliance on **3090** with site agent + optional cameras; cellular uplink for inventory sync and remote operator view — credentials and LAN I/O stay on-site.

Deploy runbook: `CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md` · Architecture detail: `Phase1-SaaS-Architecture-Costs.md`

Part 3 — Capacity limits @ wastewater fleet

Platform maximums (Phase 1 single stack)

Resource	Phase 1 max	Wastewater @ 500 assets	Headroom
Devices per tenant	1,000	500	50%
Sites / locations	1,000	~120 contractors + 32 WWTP	87%
Tags (MOOREVIEW_MAX_TAGS)	70,912 (cloud default)	~ 7,500 est.	89%
MQTT connections	~2,000	532 (500 + 32)	73%
Hot Mongo (7-day)	15 GB base plan	~ 6.5 GB	fits base tier

Tag budget by asset type

Asset type	Typical tags/site	PdM pens	Sample rate
Duplex lift station	18 – 28	Pump runtime, starts, amp imbalance	5 s (PdM)
ATU — residential	8 – 12	Blower, UV hours, high water	5 – 30 s
ATU — commercial	12 – 18	Blower, UV, pumps, compliance	5 s
WWTP head-end (appliance)	40 – 120	Blower, clarifier, pumps (local)	Rollups to cloud

500-asset tag estimate: (120 lifts × 22) + (250 ATU res × 10) + (130 ATU com × 15) ≈ **2,640 + 2,500 + 1,950 = ~7,090 tags** — set `MOOREVIEW_MAX_TAGS=8192` for wastewater tenant.

Data ingest @ 500 field assets

Lift stations run **higher-rate PdM** (cloud storage index **1.4×** per Infrastructure Projections). ATU sites are moderate.

Layer	500 assets × 50 MB/mo (weighted avg)	32 WWTP × ~100 MB/mo	Combined
Monthly ingest	~25 GB	~3.2 GB	~28 GB
Hot Mongo (7-day)	~5.8 GB	~0.7 GB	~6.5 GB
Archive Year 1	~300 GB	~38 GB	~338 GB

Archive volume: start **200 GB** (\$20/mo), expand to **~350 GB** (~\$35/mo) by month 12.

Scale triggers (wastewater fleet)

Signal	Threshold	Action
Fleet devices	> 1,000	Second MQTT broker or tenant shard
Tags	> 8,192 per tenant	Raise <code>MOOREVIEW_MAX_TAGS</code> ; scale Mongo compute
Archive	> 500 GB	Expand block volume; evaluate object storage
Lift PdM ingest	> 40 GB/mo fleet	Step SaaS to 8 GB droplet; Mongo 50 GB tier
WWTP appliances	> 50 head-ends	Dedicated site-agent relay or regional MQTT

Part 4 — Financial model

3-year revenue (wastewater vertical)

Directional model — **wastewater-focused** channel (ACE + utility + OEM). Aligns with portfolio wastewater benchmark.

Metric	Year 1	Year 2	Year 3
New assets installed	~230	~150	~120
Cumulative assets	~230	~380	~500
One-time (install + commissioning)	~\$460K	~\$375K	~\$300K
Renewal + cell (in-year + ARR build)	~\$90K	~\$365K	~\$600K
Total revenue	~\$550K	~\$1.74M	~\$3.9M
End-of-year renewal ARR	~\$55K	~\$175K	~\$400K

Year 2–3 assume 10–15% fleet renewal discount on operators above 50 sites; WWTP head-ends at commercial tier ACV.

Operating P&L — infrastructure + overhead

Line	Year 1	Year 2	Year 3
Total revenue	~\$550K	~\$1.74M	~\$3.9M
Infrastructure (cloud + cell cost)	~\$9K	~\$15K	~\$19K
Overhead (OH)	~\$6K**	~\$122K**	~\$273K**
Net after infra + OH	~\$535K	~\$1.60M	~\$3.61M
OH as % of revenue	~1.1%	7.0%	7.0%

OH policy

Period	Rule	Y1 example (~230 assets)	Y2 example	Y3 example
Year 1	\$3 / device / mo	170 avg fleet months → ~\$6K/yr**	—	—
Year 2+	7% of total revenue	—	7% × \$1.74M = ~\$122K	7% × \$3.9M = ~\$273K

OH covers G&A, support, billing, platform ops, and sales overhead — **not** field install labor or hardware COGS (those sit in install margin). Year 1 per-device OH **declines as a rate** once revenue scales; from Year 2 the model uses a **fixed 7% of total** (install + renewal + cell).

Year 1 EOY run-rate check: 230 devices × \$3 × 12 = **\$8,280/yr** if fleet were flat all year.

Year-1 site mix (wastewater launch)

Segment	Tier	Y1 target	Y1 revenue (list)	Y1 recurring added
Lift station — duplex	Commercial	40	\$113,440	\$13,440
ATU — residential	Residential	100	\$175,200	\$25,200
ATU — commercial	Commercial	60	\$170,160	\$20,160
WWTP / package plant head-end	Commercial	8	\$22,688	\$2,688
Wells & pumps (adjacent)	Commercial	22	\$62,392	\$7,392
Total		~230	~\$543,880	~\$68,880 ARR

Rounded to ~\$550K year-1 total with in-year renewal recognition and OEM NRE.

Steady-state fleet @ 500 assets (Year 3 end)

Reference mix for Phase 1 capacity and infra cost model:

Segment	Count	Renewal/mo	Monthly renewal
Lift station — duplex	120	\$25	\$3,000
ATU — residential	250	\$18	\$4,500
ATU — commercial	100	\$25	\$2,500
WWTP head-end (no per-asset renewal)*	32	—	—
Total billable assets	470		\$10,000

WWTP head-ends billed as commercial sites (\$25/mo) in operator contracts; 32 appliances share tenant fleet view.
Full 500-device MQTT count = 470 field + 32 head-end + overhead.

With 32 WWTP @ \$25/mo: total renewal ~~\$10,800/mo~~ (\$129,600/yr).

Renewal rate sensitivity — \$20 / \$25 / \$30 per month

Income-side comparison at **Year 3 steady state** (~500 billable field assets + 32 WWTP head-ends).

Infrastructure held at Phase 1 actuals (~~\$1,571/mo~~). ~~OH: Year 1 = \$3/device/mo; Year 2+ = 7% of total revenue~~
~~— sensitivity tables show 7% applied to recurring subtotal; full-company OH at Y3 total (\$3.9M) = ~\$273K/yr.~~

Uniform commercial rate (500 assets × rate)

Assumes all field assets priced at the same monthly renewal (sensitivity band around list **\$25/mo** commercial tier).

Renewal rate	Monthly renewal	Annual renewal	+ Cell (\$3 × 500)	Total recurring/mo	Infra	OH (Y1 \$3/dev)	OH (Y2+ 7%)*	Net/mo Y1	**Net Y2+*
\$20 / mo	\$10,000	\$120,000	\$1,500	\$11,500	— \$1,571	— \$1,500	—\$805	\$7,929	\$8,614
\$25 / mo (list)	\$12,500	\$150,000	\$1,500	\$14,000	— \$1,571	— \$1,500	—\$980	\$10,429	\$10,919
\$30 / mo	\$15,000	\$180,000	\$1,500	\$16,500	— \$1,571	— \$1,500	— \$1,155	\$12,429	\$12,584

*7% × monthly recurring subtotal (\$11,500 / \$14,000 / \$16,500). +Full Y3 company OH on \$3.9M total ≈ **\$22,750/mo** (includes install + renewal mix — not recurring-only).

Annual net (recurring subtotal only, uniform 500 @ \$25/mo):

	Infra	OH Y1 (\$3/dev)	OH Y2+ (7% recurring)	Net Y1	Net Y2+
\$20/mo	—\$18,855	—\$18,000	—\$9,660	\$91,485	\$99,825
\$25/mo	—\$18,855	—\$18,000	—\$11,760	\$119,385	\$125,625
\$30/mo	—\$18,855	—\$18,000	—\$13,860	\$147,285	\$151,425

Y1 OH: $500 \times \$3 \times 12 = \$18,000/\text{yr}$ at full fleet (Y3 steady state). During actual Y1 (230 assets), OH ~ ~~***~~\$6K/yr**.

Per asset (renewal + cell, after infra + OH allocated @ 500 fleet):

Rate	Gross/asset/mo	Infra/asset	OH Y1 (\$3)	OH Y2+ (7%)*	Net Y1/asset	Net Y2+/asset
\$20	\$23.00	~\$3.14	\$3.00	~\$1.61	~\$16.86	~\$17.25
\$25	\$28.00	~\$3.14	\$3.00	~\$1.96	~\$21.86	~\$21.90
\$30	\$33.00	~\$3.14	\$3.00	~\$2.31	~\$26.86	~\$26.55

*7% of recurring subtotal ÷ 500 assets. Y1 OH is **higher per asset** (\$3) than Y2+ recurring-only allocation until total revenue includes install mix.

Blended fleet mix (planning default)

Steady-state mix from table above: **250 ATU res @ \$18 · 250 commercial @ rate** (lift + ATU com + wells). Shows effect of residential tier anchoring the blend.

Commercial rate	Blended renewal/mo	Total recurring/mo	Infra	OH Y1	OH Y2+ (7%)	Net Y1/mo	Net Y2+/mo
\$20 / mo	\$9,500	\$11,000	– \$1,571	– \$1,500	–\$770	\$6,959	\$7,659
\$25 / mo	\$10,750	\$12,250	– \$1,571	– \$1,500	–\$858	\$8,679	\$9,321
\$30 / mo	\$12,000	\$13,500	– \$1,571	– \$1,500	–\$945	\$10,429	\$10,984

Blended: $(250 \times \text{commercial rate}) + (250 \times \$18)$.

End-of-year renewal ARR — rate comparison

Rate scenario	Y1 (~230 assets)*	Y2 (~380 assets)*	Y3 (~500 assets)
\$20 / mo (uniform)	~\$44K ARR	~\$91K ARR	~\$120K ARR
\$25 / mo (uniform)	~\$55K ARR	~\$114K ARR	~\$150K ARR
\$30 / mo (uniform)	~\$66K ARR	~\$137K ARR	~\$180K ARR
List blend (\$18 res / \$25 com)	~\$55K ARR	~\$175K ARR	~\$129 – 400K ARR†

*Uniform rate applied to all billable assets for sensitivity; actual Y1 mix includes one-time install revenue (~~\$460K Y1~~).
†Y3 ARR range: blend table (\$129K renewal only) vs plan total (~\$400K) includes WWTP head-ends, cell, and fleet discounts.

Total revenue impact (renewal portion only, Y3)

Hold Y3 **one-time install (~\$300K)** constant; vary renewal + cell on **500-asset** base:

Rate	Y3 renewal + cell revenue	Y3 one-time	Y3 total (directional)
\$20 / mo	~\$138K	~\$300K	~\$438K
\$25 / mo	~\$168K	~\$300K	~\$468K
\$30 / mo	~\$198K	~\$300K	~\$498K

*Full plan ~**\$3.9M Y3** includes Y2 carry-forward installs, OEM NRE, and in-year ramp recognition — not just steady-state renewal on 500 assets. Use this table for **renewal pricing decisions**; use Part 4 3-year model for **channel ramp totals*.

Pricing takeaway

	\$20/mo	\$25/mo (list)	\$30/mo
vs autodialer (~\$50–100/mo effective)	Still premium capability	List anchor	Utility/municipal band
Net margin @ 500 fleet (Y1, after infra + \$3 OH)	~ 69%	~ 75%	~ 75%
Net margin @ 500 fleet (Y2+, after infra + 7% OH)*	~ 75%	~ 78%	~ 81%
Increment vs \$25 (500 assets)	–\$2,500/mo renewal	—	+\$2,500/mo renewal
Break-even infra + Y1 OH (500 fleet)	~ 200 assets @ \$20	~165 @ \$25	~140 @ \$30

*7% OH on recurring subtotal only; company-wide Y2+ OH on full \$1.74M / \$3.9M revenue is higher in absolute dollars.

\$25/mo is list commercial (Pricing Guide v2.4). **\$20/mo** supports volume fleet / utility pilot (10–15% discount band starts at 50 sites). **\$30/mo** fits municipal / compliance-heavy lift stations without changing Phase 1 infra.

Part 5 — Infrastructure & cellular cost (2026)

Actual carrier costs (planning basis)

Fleet class	SIM	Data plan	Cost/asset/mo	List to customer
Field assets (lift, ATU, IoT-Link)	\$2.00	\$0.50 / 50 MB	\$2.50	\$3.00
WWTP head-end appliances (cameras, site agent)	\$2.00	\$5.00 / 1 GB	\$7.00	Pass-through or bundled

50 MB/mo per lift/ATU is sufficient for 5 s Parc telemetry + alarm events. 1 GB/mo supports site-agent heartbeat, inventory sync, and occasional camera proxy — not continuous HD streaming.

Cloud infrastructure (Phase 1 @ 500 assets)

Component	Monthly	Annual
SaaS 4 GB droplet	\$24	\$288
MQTT 2 GB droplet	\$12	\$144
Archive 2 GB droplet	\$12	\$144
Archive volume (~338 GB Y1 avg)	~\$34	~\$408
Managed Mongo (15 GB base)	\$15.23	\$183
Cloud subtotal	~\$97	~\$1,167

Cellular @ 500 + 32

Line	Calculation	Monthly	Annual
Field fleet (500 × \$2.50)	SIM + 50 MB	\$1,250	\$15,000
WWTP head-ends (32 × \$7.00)	SIM + 1 GB	\$224	\$2,688
Cellular subtotal		\$1,474	\$17,688

Combined P&L — infrastructure + overhead (Year 3 steady state)

	Monthly	Annual	% of renewal
Renewal revenue (500 assets, list)	\$10,800	\$129,600	100%
DO Phase 1 cloud	−\$97	−\$1,167	0.9%
Cellular — field (cost)	−\$1,250	−\$15,000	11.6%
Cellular — WWTP (cost)	−\$224	−\$2,688	2.1%
Total infrastructure	−\$1,571	−\$18,855	14.5%
Overhead — Y1 (\$3 × 500 devices)	−\$1,500	−\$18,000	13.9%
Overhead — Y2+ (7% of total revenue)	−\$22,750	−\$273,000	—†
Net after infra + OH (Y1 rate)	\$7,729	\$92,745	71.5%
Net after infra + OH (Y2+ company)	—‡	~\$3.61M	—‡

†Y2+ OH is **7% of ~\$3.9M total revenue** (install + renewal), not renewal-only. ‡Company net shown in Part 4 3-year P&L; recurring-only steady-state net @ \$25 ≈ **\$10,949/mo** (7% on \$14K recurring).

Cell pass-through margin

	Monthly
Cell billed (500 × \$3 list)	\$1,500
Cell cost (500 × \$2.50)	−\$1,250
Cell margin	\$250
WWTP cell (32 × \$7 cost; often bundled in site fee)	−\$224

Per-asset economics (commercial lift station)

	Per asset/mo	Per asset/yr
Renewal revenue	\$25.00	\$300.00
Cell billed (list)	\$3.00	\$36.00
Cell cost	\$2.50	\$30.00
Cloud infra (allocated, 500 fleet)	~\$0.19	~\$2.33
WWTP uplink (allocated)	~\$0.04	~\$0.54
Overhead Y1 (\$3/device)	\$3.00	\$36.00
Overhead Y2+ (7% of recurring)*	~\$1.96	~\$23.52
Net after infra + OH (Y1)	~\$19.27	~\$231.13
Net after infra + OH (Y2+ recurring)	~\$21.90	~\$262.80

Gross margin on renewal + cell (list, Y1): $(\$28 - \$2.73 \text{ infra} - \$3 \text{ OH}) / \$28 \approx 79\%$ before install labor and hardware COGS.

*7% × \$28 recurring subtotal per asset ≈ \$1.96/mo; company-wide 7% on total revenue differs when install is included.

Infrastructure + overhead cost by year (ramp)

Year	Cumulative assets	Cloud/mo	Cell/mo	Infra/mo	OH/mo	OH rule	Total opex/mo
Y1	~230	~\$93	~\$687	~\$780	~\$510	\$3 × ~170 avg fleet	~\$1,290
Y2	~380	~\$95	~\$1,118	~\$1,213	~\$10,150	7% × \$1.74M	~\$11,363
Y3	~500	~\$97	~\$1,474	~\$1,571	~\$22,750	7% × \$3.9M	~\$24,321

Y1 OH uses average fleet (~170 device-months) → ~\$510/mo. Y2–Y3 OH from **7% of annual total revenue** ÷ 12.

Phase 1 stack is **largely fixed** from launch — margin improves with fleet density until archive/Mongo tier step-up (~600+ high-rate lift assets).

Break-even (infrastructure only)

Cost covered	Assets @ \$25/mo renewal
Cloud stack alone (~\$97/mo)	~4 assets
Cloud + field cell (~\$1,347/mo)	~54 assets
Full stack incl. 32 WWTP (~\$1,571/mo)	~63 assets

Part 6 — Unit economics & targets

Metric	Target
Renewal gross margin (after cloud + cell + OH)	70 – 80% (Y1) · 75 – 85% (Y2+ recurring)
Hardware + commissioning margin	35 – 45%
Blended year-1 payback on install COGS	Under 14 months
Net revenue retention (fleet operators)	> 110%
Logo churn (annual)	< 5%
LTV (commercial lift, 7 yr)	\$2,500 + (\$336 × 7) ≈ \$4,852

LTV:CAC target > 3:1 on contractor-led installs. ACE channel reduces CAC vs cold national sales.

Part 7 — Go-to-market

Phase roadmap

Phase	Assets	Focus	Cloud
Launch (Y1)	~230	ACE reference fleet; lift + ATU routes	Deploy Phase 1; 10 GB Mongo
Growth (Y2)	~380	Utility pilots; OEM panel embed	Archive expand 200→300 GB
Scale (Y3)	~500	Fleet discounts; municipal RFP	Phase 1 at capacity; plan ingest shard

Sales channels (wastewater)

Channel	Y1 mix	Advantage
---------	--------	-----------

Purple Standard / ACE	40 – 50%	Install crews, brand, first 500 sites
Contractor monitoring contracts	30 – 40%	Attach on PM routes
OEM / panel embed	10 – 15%	ATS panel + provider Parc
Utility / municipal	5 – 10%	Longer cycle; 13 mo compliance archive

Compliance & retention drivers

- 13-month historian archive on lift/ATU (regulatory route documentation)
- PdM → proactive CMMS — service due before inspection failure
- Portable .est — contractor owns project; no vendor lock-in on recommission

Part 8 — Risk & scale path

Risk	Mitigation
Cellular outage at lift station	Local alarm logic on Opta; spool on IoT-Link; SSO dialer as last line
MQTT broker load	Dedicated Phase 1 MQTT droplet; lock pilot IPs; scale at 2K connections
Archive growth (lift PdM)	7-day hot → zstd; expand block volume \$0.10/GiB
Tenant tag overflow	Size MOOREVIEW_MAX_TAGS=8192 for wastewater; monitor dashboard
National scale (>5K assets)	VM split: ingestion · alarms · GUI · AI (ARCHITECTURE.md)

Part 9 — Comparison to Product Market Entry (portfolio)

This vertical plan vs **MooreVIEW-Product-Market-Entry.md v1.5** (full portfolio launch).

Revenue trajectory

Metric	Wastewater (this plan)	Portfolio market entry
Y1 total revenue	~\$550K	~\$550K – \$600K
Y2 total revenue	~\$1.74M	~\$1.5M
Y3 total revenue	~\$3.9M	~\$3.5M
Y1 assets/sites	~230 WW	237 mixed (95 WW = 40%)
Y3 cumulative	~500 WW	~1,487 all verticals
Y3 renewal ARR	~\$400K	~\$780K

Wastewater-only **concentrates revenue** in fewer assets with **higher Y2–Y3 totals** than the blended portfolio — because ACE/Purple Standard channel and utility pilots weight lift + ATU before pools/HVAC scale.

Infrastructure & Phase 1 cloud

Dimension	Wastewater @ 500	Portfolio @ 1,487 sites
Deployment model	Cloud-centric (IoT-Link)	Mostly cloud-centric + 2 ALF Y1
Phase 1 device load	500 MQTT — at stack capacity	~1,487 sparse — same stack OK
Cloud tags (est.)	~7,090	~9,000 + ALF rollups
Hot Mongo (7 d)	~6.5 GB	~15 – 60 GB (Y1 band)
Dominant opex	Cellular \$1,250/mo	Cellular + archive growth
Infra % of renewal	~15% @ WW list pricing	~15 – 31% by mix
Cloud stack cost	~\$97/mo	~\$97 – 145/mo (Y1 – Y3)

Pricing alignment

Both documents use **Pricing Guide v2.4** list: commercial **\$2,836** Y1 · **\$336/yr** recurring with cell. This plan adds **actual carrier cost** (\$2.50/mo field) for margin planning; Product Market Entry uses **\$3/mo list** cell line item.

What portfolio includes that WW omits

Vertical	Y1 sites (portfolio)	In WW plan?
Pools (res + com)	45	No
HVAC (res + com)	70	No
Convenience store	10	No
ALF / campus	2	No (see ALF scaling in Infrastructure Projections Part 10)
Wells (adjacent)	15	Partial (22 in WW Y1 mix)

Strategic positioning

	Wastewater vertical	Portfolio
Beachhead	ACE Septic · lift + ATU routes	Lift + c-store + pools + HVAC
Phase 1 stress test	500 devices on one MQTT broker	Device count higher; telemetry sparser
5-yr ARR ceiling	\$15 – 25M (WW SAM)	\$50 – 120M (all verticals)
When to use this doc	Utility RFP · ACE fleet · WWTP OEM	Investor · multi-vertical GTM

Appendix — assumptions

Item	Value
Fleet reference	500 field assets + 32 WWTP head-ends
Field cellular cost	\$2/SIM + \$0.50 per 50 MB/mo
WWTP cellular cost	\$2/SIM + \$5 per 1 GB/mo
Field cellular list	\$3/mo (Pricing Guide v2.4)
Cloud topology	Phase 1 NYC1 (SaaS + MQTT + Archive + Mongo)
Revenue benchmarks	Wastewater vertical plan Y1–Y3
Tag estimate	~7,090 @ 500 assets; cap 8,192
Ingest	50 MB/mo/field asset; ~100 MB/mo/WWTP head-end
DO pricing	Aug 2026 list
Overhead Y1	\$3 / device / mo on cumulative monitored assets
Overhead Y2+	7% of total revenue (install + renewal + cell)

Disclaimer: Directional business plan for capacity, pricing, and opex review. Not a financial forecast, hardware quote, or SLA commitment.

Related documents

Document	Purpose
Phase1-SaaS-Architecture-Costs.md	Phase 1 topology + generic 500-device cost model
D0-Phase1-Storage-Cost-Review.md	Storage-only deep dive
MooreVIEW-Pricing-Guide.md	List MSRP v2.4
MooreVIEW-Product-Market-Entry.md	Portfolio launch model v1.5 (comparison Part 9)
MooreVIEW-Infrastructure-Projections.md	5-year scaling · Part 9 Phase 1 · Part 10 ALF
CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md	Production deploy runbook

Wastewater Business Plan v1.3 — mooreVIEW / The Purple Standard.

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